Copper: Fact Sheet for Consumers (original version)

# Copper: Fact Sheet for Consumers

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## What is copper and what does it do?

Copper is a mineral that you need to stay healthy. Your body uses copper to carry out many important functions, including making energy, connective tissues, and blood vessels. Copper also helps maintain the nervous and immune systems and activates genes. Your body also needs copper for brain development.

## How much copper do I need?

The amount of copper you need each day depends on your age. Average daily recommended amounts are listed below in micrograms (mcg).

| Life Stage | Recommended Amount |
|---|---|
| Birth to 6 months | 200 mcg |
| Infants 7–12 months | 220 mcg |
| Children 1–3 years | 340 mcg |
| Children 4–8 years | 440 mcg |
| Children 9–13 years | 700 mcg |
| Teens 14–18 years | 890 mcg |
| Adults 19 years and older | 900 mcg |
| Pregnant teens and women | 1,000 mcg |
| Breastfeeding teens and women | 1,300 mcg |

## What foods provide copper?

Many foods contain copper. You can get recommended amounts of copper by eating a variety of foods, including the following:

- Beef liver and shellfish such as oysters
- Nuts (such as cashews), seeds (such as sesame and sunflower), and chocolate
- Wheat-bran cereals and whole-grain products

- Potatoes, mushrooms, avocados, chickpeas, and tofu

# What kinds of copper dietary supplements are available?

Copper is available in many multivitamin/mineral supplements, in supplements that contain only copper, and in other dietary supplements. Copper in dietary supplements is often in the forms of cupric oxide, cupric sulfate, copper amino acid chelates, and copper gluconate. It is not known whether one form of copper is better than another.

# Am I getting enough copper?

Most people get enough copper from the foods they eat. However, certain groups of people are more likely than others to have trouble getting enough copper:

- People with celiac disease
- People with Menkes disease, a rare genetic disorder
- People taking high doses of zinc supplements, which can interfere with the ability to absorb copper and could lead to copper deficiency

# What happens if I don’t get enough copper?

Copper deficiency is rare in the United States. Copper deficiency can cause extreme tiredness, lightened patches of skin, high levels of cholesterol in the blood, and connective tissue disorders affecting the ligaments and skin. Other effects of copper deficiency are weak and brittle bones, loss of balance and coordination, and increased risk of infection.

# What are some effects of copper on health?

Scientists are studying copper to understand how it affects health. Here are several examples of what this research has shown.

### Cardiovascular disease

Studies looking at the effect of copper intake on heart disease have had mixed results. More research is needed to understand whether getting more copper from the diet or supplements might raise or lower the risk of cardiovascular disease.

### Alzheimer’s disease

Some research shows that people with higher levels of copper in their blood have a lower risk of Alzheimer’s disease. Other research, however, shows that high amounts might increase Alzheimer’s disease risk. More research is needed to determine whether high or low levels of copper affect the risk of developing Alzheimer’s disease. Research is also needed to find out whether dietary supplements that contain copper could affect the risk of Alzheimer’s disease or its symptoms.

# Can copper be harmful?

Yes, copper can be harmful if you get too much. Getting too much copper on a regular basis can cause liver damage, abdominal pain, cramps, nausea, diarrhea, and vomiting. Copper toxicity is rare in healthy individuals. However, it can occur in people with Wilson’s disease, a rare genetic disorder. It can also occur if copper-containing water pipes leach copper into drinking water in your home or workplace.

The daily upper limits for copper include intakes from all sources—food, beverages, and supplements—and are listed below in micrograms (mcg).

| Ages | Upper Limit |
|---|---|
| Birth to 12 months | Not established |
| Children 1–3 years | 1,000 mcg |
| Children 4–8 years | 3,000 mcg |
| Children 9–13 years | 5,000 mcg |
| Teens 14–18 years | 8,000 mcg |
| Adults | 10,000 mcg |

## Does copper interact with medications or other dietary supplements?

Copper is not known to interact with any medications. However, it's always important to tell your doctor, pharmacist, and other health care providers about any dietary supplements and prescription or over-the-counter medicines you take. They can tell you if the dietary supplements might interact with your medicines or if the medicines might interfere with how your body absorbs, uses, or breaks down nutrients such as copper.

## Copper and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the *Dietary Guidelines for Americans*.

## Where can I find out more about copper?

- For more information on copper
  - Office of Dietary Supplements (ODS) Health Professional Fact Sheet on Copper
  - MedlinePlus, Copper in diet
- For more information on food sources of copper
  - ODS Health Professional Fact Sheet on Copper
  - USDA's FoodData Central
  - USDA, Copper content of selected foods
- For more advice on choosing dietary supplements
  - ODS Frequently Asked Questions: Which brand(s) of dietary supplements should I purchase?
- For information about building a healthy dietary pattern
  - Dietary Guidelines for Americans

## Glossary

absorption
: In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

Alzheimer's disease
: A brain disease in which thinking, memory, and reasoning ability is slowly destroyed. In advanced stages, an affected person becomes disoriented and confused, has mood and behavior changes, and has difficulty talking, walking, and swallowing. Alzheimer's disease is progressive, irreversible, and incurable.

amino acid
: A chemical building block of protein.

blood vessel
: A tube through which blood circulates in the body. Blood vessels include a network of arteries, arterioles, capillaries, venules, and veins.

celiac disease
: An autoimmune disorder in which eating gluten (a protein found in wheat, rye, barley, and possibly oats) causes the immune system to damage the small intestine, making it unable to absorb nutrients. It is a genetic disease that sometimes becomes active for the first time after surgery, pregnancy, childbirth, viral infection, or extreme stress. Also called sprue.

cholesterol
: A substance found throughout the body. It is made by the liver and is an important component of cells. Cholesterol is also used to make hormones, bile acid, and vitamin D. Foods that come from animals contain cholesterol, including eggs, dairy products, meat, poultry and fish. High blood levels of cholesterol increase a person's chance (risk) of developing atherosclerosis and heart disease.

connective tissue
: Cells that work together to protect and support the body's muscles, joints, organs, skin, and other tissues. Examples of connective tissue include cartilage, fat, blood, and bone.

copper
: In nutrition, a mineral the body needs (along with iron) to make red blood cells. Copper also helps keep the immune system, blood vessels, nerves, and bones healthy. Copper is found in some foods, including oysters and other shellfish, whole grains, beans, nuts, potatoes, organ meats, dark leafy greens, and dried fruits.

deficiency
: An amount that is not enough; a shortage.

diarrhea
: Loose, watery stools.

dietary fiber
: A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans
: Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement
: A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

disorder
: In medicine, a disturbance of normal functioning of the mind or body. Disorders may be caused by genetic factors, disease, or trauma.

dose
: The amount of medicine or other substance taken at one time or over a specific period of time.

fortified
: When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

gene
: The functional and physical unit of heredity passed from parent to offspring. Genes are pieces of DNA, and most genes contain the information for making a specific protein.

genetic disorder
: A disease or disorder caused by an alteration or variation (mutation) in a gene or group of genes in the cells of an individual. Examples of genetic disorders include breast cancer, cystic fibrosis, Parkinson's disease, and celiac disease. They can be inherited or can occur without a known cause.

immune system
: A group of organs and cells that defends the body against infection, disease, and altered (mutated) cells. It includes the thymus, spleen, lymphatic system (lymph nodes and lymph vessels), bone marrow, tonsils, and white blood cells.

infant
: A child younger than 12 months old.

infection
: The invasion and spread of germs in the body. The germs may be bacteria, viruses, yeast, or fungi.

interaction
: A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

liver
: A large organ located in the right upper abdomen. It stores nutrients that come from food, makes chemicals needed by the body, and breaks down some medicines and harmful substances so they can be removed from the body.

microgram
: $\mu$g or mcg. A unit of weight in the metric system equal to one millionth of a gram. (A gram is approximately one-thirtieth of an ounce.)

mineral
: In nutrition, an inorganic substance found in the earth that is required to maintain health.

nausea
: The uneasy feeling of having an urge to throw up (vomit).

nutrient
: A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

Office of Dietary Supplements
: ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

pharmacist
: A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

prescription
: A written order from a health care provider for medicine, therapy, or tests.

risk
: The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

supplement
: A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

symptom
: A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

tissue
: A group or layer of cells in a living organism that work together to perform a specific function.

toxicity
: The degree to which something is poisonous (toxic).

upper limit
: UL. The largest daily intake of a nutrient considered safe for most people. Taking more than the UL is not recommended and may be harmful. The UL for each nutrient is set by the Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine. For example, the UL for vitamin A is 3,000 micrograms/day. Women who consume more than this amount every day shortly before or during pregnancy have an increased chance (risk) of having a baby with a birth defect. Also called the tolerable upper intake level.

US Department of Agriculture

USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US public and private lands, and providing financial programs to improve the economy and quality of rural American life.

vitamin
: A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

zinc
: A mineral found in most cells of the body. It helps enzymes work properly, helps maintain a healthy immune system, helps maintain the senses of taste and smell, and is needed for wound healing, making DNA, and normal growth and development during pregnancy, childhood, and adolescence. Zinc is found in some foods, including oysters, red meat, poultry, beans, nuts, certain seafood, whole grains, fortified breakfast cereals, and dairy products.

**Updated:** October 18, 2022